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OL	T	H	W	R
Sunny	Hot	High	Weak	No
Sunny	Mild	High	Strong	No
Cloudy	Hot	High	Weak	Yes
Rainy	Mild	High	Weak	Yes
Rainy	Cool	Normal	Weak	Yes

OL = Sunny

$\Rightarrow \frac{0}{2}\log_2\left(\frac{0}{2}\right) - \frac{2}{2}\log_2\left(\frac{2}{2}\right)$

$\Rightarrow 0$

Sunny = 2
Cloudy = 1
Rainy = 2
Total = 5

Cloudy =

$= \frac{1}{1}\log_2\left(\frac{1}{1}\right) - \frac{0}{1}\log_2\left(\frac{0}{1}\right)$

$\Rightarrow 0$

Rainy =

$= \frac{2}{2}\log_2\left(\frac{2}{2}\right) - \frac{0}{2}\log_2\left(\frac{0}{2}\right)$

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$$R = -\frac{3}{5}\left[\log_2\left(\frac{3}{5}\right)\right] - \frac{2}{5}\left[\log_2\left(\frac{2}{5}\right)\right]$$

$\Rightarrow$ ~~0.08~~ 0.97

The weighted entropy is $= \frac{2}{5}(0) + \frac{3}{5}(0)$

$\Rightarrow 0$

IG(O) = 0.97

T,

Hot

$\Rightarrow -\frac{1}{2}\log_2\left(\frac{1}{2}\right) - \frac{1}{2}\log_2\left(\frac{1}{2}\right)$

$\Rightarrow -1$

Hot = 2
Mild = 2
Cool = 1
Total = 5

Mild,

$\Rightarrow -\frac{1}{2}\log_2\left(\frac{1}{2}\right) - \frac{1}{2}\log_2\left(\frac{1}{2}\right)$

$\Rightarrow 1$

Cool =

$\Rightarrow \frac{1}{1}$ lo

$\Rightarrow$ 0

IG =

IG.(T

H,

High

$\Rightarrow -\frac{2}{}$

$\Rightarrow$

Normal

$\Rightarrow$

$\Rightarrow$

Weigh

Cool =

$$= \frac{1}{1}\log_2\left(\frac{1}{1}\right) - \frac{0}{1}\log_2\left(\frac{0}{1}\right)$$

$$\Rightarrow 0$$

IG = Weighted entropy $= \frac{2}{5}(1) + \frac{2}{5}(1) + \frac{1}{5}(0)$

$= 0.8$

$IG.(T) = 0.97 - 0.8 = 0.17$

H,

High = 4
Normal = 1

High

$$\Rightarrow -\frac{2}{4}\log_2\left(\frac{2}{4}\right) - \frac{2}{4}\log_2\left(\frac{2}{4}\right)$$

$\Rightarrow 1$

Normal;

$$\Rightarrow \frac{1}{1}\log_2\left(\frac{1}{1}\right) - \frac{0}{1}\log_2\left(\frac{0}{1}\right)$$

$\Rightarrow 0$

Weighted entropy $= \frac{4}{5}(1) + \frac{1}{5}(0) = 0.8$